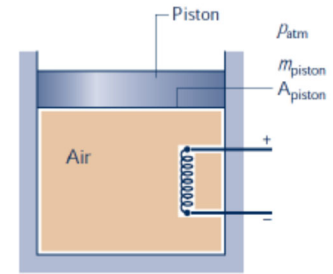


You do not need to include assumptions and basic equations for HW-3.

You do not need to include system diagram, assumptions, and basic equations for HW-4.

HW – 3(i)

A closed and externally well-insulated piston-cylinder device contains air. An electric resistance heater adds energy to the air causing it to expand in the cylinder and moving the piston.

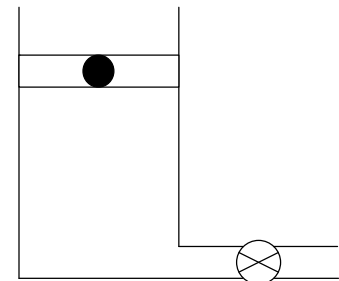


Show mass and energy flows at the boundary for the following systems:

- (a) Only air inside the cylinder
- (b) Only electric resistance heater
- (c) Air and electric resistance heater

HW – 3(ii)

High-pressure gas from a large supply line enters into a well-insulated piston-cylinder device when the valve is open and the gas pushes the piston upward.



Show mass and energy flows at the boundary for gas in the cylinder.

HW – 3(iii)

An automobile radiator is a rigid heat exchanger that cools the cooling water in an engine. Hot water from the engine enters into the finned tubes at the top and cooled water returns to the engine from the bottom. Air flow across the finned tubes cools the water flowing through the tubes.



Show mass and energy flows at the boundary for the following systems:

- (a) Only water flowing through finned tubes in the radiator
- (b) Air and water in the radiator

HW – 4(i)

A closed rigid tank of volume 2 m^3 contains 5 kg of a certain gas at an absolute pressure of 100 kPa and a temperature of 27°C .

- (a) Identify two extensive and four intensive properties.
- (b) List numerical values of the properties identified in (a) with their units.

HW – 4(ii)

Determine whether the following statements are true or false with appropriate justification for your choice.

- (a) Chemical composition of a closed system can change.
- (b) Work is relevant to a particular state of a system.
- (c) Heat transfer between system and surrounding is zero for a system at thermal equilibrium.
- (d) Pressure always remains constant during a quasi-equilibrium process.

HW – 4(iii)

A gas ($m = 2 \text{ kg}$) is contained in a closed, rigid tank. Heat transfer of 30 kJ occurs to the gas uniformly over a period of 1 minute .

- (a) Calculate the specific heat transfer during the process, in kJ/kg .
- (b) Find the rate of heat transfer during the process, in kW .